

GradeSense Assessment - Answer Sheet

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Total marks: 15

Answer 1. (Science)

A simple electric circuit is a closed path which allows the current to flow through it. The battery is the source of energy and it provides the potential difference which pushes the current around the loop. The switch is used to open and close the circuit. When the switch is closed the path is complete and current flows through the bulb and the resistor, and when the switch is open the path is broken so no current can flow.

In my circuit the battery, the switch, the resistor, the bulb and the ammeter are all joined one after the other in series. The ammeter is connected in series because it has to measure the current that is passing through the circuit. The voltmeter is also connected in series, just after the bulb, so that it can measure the voltage of the bulb.

The current comes out from the positive terminal of the battery, goes through all the components of the outside circuit and returns back into the negative terminal.

The resistance decides how much current will flow in the circuit. If we increase the resistance then the current also increases, because a bigger resistor pushes more current through the wire. This is the reason a rheostat is used when we want to make the bulb glow brighter.

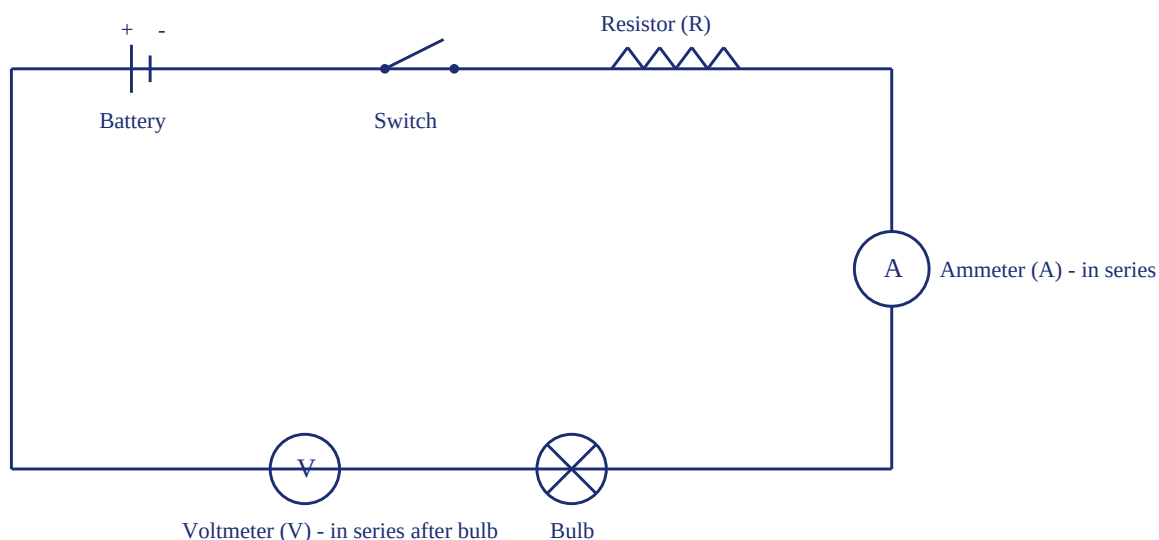


Fig. 1 - Circuit diagram of a simple electric circuit

Answer 2. (English)

In today's world technology has made information very easy to get. Anything a student wants to know is available on the internet within a few seconds. I believe that technology is making students more dependent on ready-made answers instead of making them better learners.

Earlier a student had to read the whole chapter and think about the question properly before writing the answer. Now most students simply search the question on their phone and copy the first answer that comes. Because of this they finish their homework very fast but they do not remember anything

after a few days. For example, many students now use a calculator even for very small sums, and later they cannot do a simple multiplication in their head. Some people say that technology is good because it gives more information.

Therefore technology is a very useful thing for students and it definitely makes them better learners, so every student should use it as much as possible.

Answer 3. (Economics)

The table shows how much quantity is demanded and how much is supplied at each different price. I have drawn the graph below by taking price on the horizontal axis and quantity on the vertical axis.

The demand curve slopes downward because when the price is low the buyers want to buy a larger quantity, and the supply curve slopes upward because the sellers are willing to sell more when the price is high. The two lines cut each other at the price of Rs. 30 and the quantity of 60 units. This is the equilibrium point of the market, because at this price the quantity demanded is exactly equal to the quantity supplied.

If the price in the market goes above the equilibrium price then there will be a shortage in the market, because the buyers will demand much more than what the sellers are ready to supply. In the same way, if the price falls below the equilibrium price there will be a surplus of goods lying unsold with the sellers.

If the cost of production increases then the producers will not be able to supply the same quantity at the old price, so the supply curve shifts to the left. The new equilibrium will then be at a higher price and also at a higher quantity, because when the price has gone up the sellers will sell more.

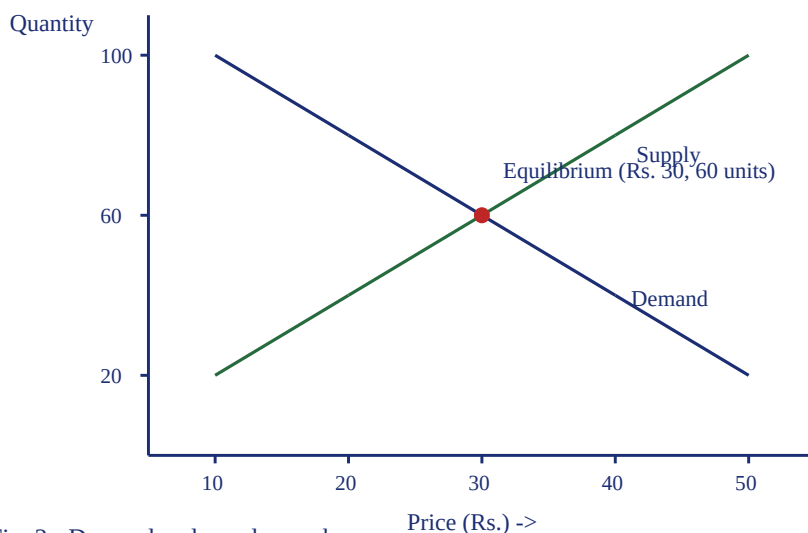


Fig. 2 - Demand and supply graph